

Section J – 1602000 Electrical	
Attachment Number	Attachment Title
J-1602000-01	Definitions and Acronyms
J-1602000-02	References and Technical Documents
J-1602000-03	Electrical Power Transmission and Distribution Systems Description and Inventory
J-1602000-04	Minimum Critical Spare Parts List

Attachment J-1602000-01 Definitions and Acronyms	
Equipment, Collateral	Encompasses built-in and large substantially affixed equipment/property that is normally acquired and installed as part of a facility project.
Equipment, Installed	Encompasses building-type equipment, built-in equipment, and large, substantially affixed equipment/property, and is normally acquired and installed as part of a facility project. Installed equipment is normally required to make a facility useful and operable. Removing such equipment would impair the usefulness, safety, or environment of the facility or the facility restoration work required after its removal, is substantial.
Equipment, Personal Property	Personal property equipment includes all equipment other than collateral equipment. Such equipment, when acquired and used in a facility or a test apparatus, can be severed and removed after erection or installation without substantial loss of value or damage thereto or to the premises where installed.
Facilities Life Cycle	A facilities life cycle is divided into four stages, requirements (planning and design), acquisition (construction and acceptance), stewardship (operations, maintenance and repair), and disposal.
Facilities Maintenance Management	The planning, prioritizing, organizing, controlling, reporting, evaluating, and adjusting of facilities maintenance operations to support the CNO/NAVFAC facilities policy and objectives and satisfy customers' facility needs. Defined by the International Facility Management Association as "the practice of coordinating the physical workplace with the people and work of the organization."
High Voltage Transmission	Transmission of electrical loads greater than 600Volts (V) and less than 24kilovolts (kV)
Integrated Maintenance Plan	IMP is a recurring state-of-the-art, reliability-centered inspection, testing, maintenance and repair program that determines best practices for managing the functions and consequences of failures of facilities equipment and system components. IMP encompasses accepted commercial practices, including reactive, preventive, predictive and proactive maintenance, into one optimal program. The IMP approach gives the Contractor full responsibility to maintain systems and equipment and perform repairs whenever necessary to ensure equipment and systems are operational and remain in a constant state of readiness.
Life-Cycle Costs	A form of economic analysis that considers the total cost of owning, operating, and maintaining a building or system over its useful life.
Low Voltage Distribution	Distribution of electrical loads less than or equal to 600 volts (V).
Medium Voltage Distribution	Distribution of electrical loads ranging from greater than 600 V up to and including 24 kilovolts (kV).
NEC	National Electrical Code
NFPA	National Fire Protection Association
Repair	Repair is the restoration of facilities or equipment to such a condition that it may be effectively utilized for its designated purposes by overhaul, reconstruction, or replacement of constituent parts or materials which have deteriorated by action of the elements or usage, and which have not been corrected through maintenance. This term also applies to replacement of the entire unit or system if beyond economical repair. The intent of repair is to have the equipment at normal working condition.
Replacement	Replacement, as a distinct work element, is confined to a program of planned replacement of a facility or its components. It may be further limited to major components such as air conditioning compressors, furnaces or hot water heaters. Replacement is performed when the equipment has reached the end of its useful life; when it no longer can perform due to degradation of its internal components and repair is no longer cost effective. Included under the replacement would be the major rebuilding of any component, since rebuilding also restores performance.

Attachment J-1602000-01
Definitions and Acronyms

Restoration	Restoration of real property to such a condition that it can be used for its intended purpose. Includes repair or replacement work to restore facilities damaged by inadequate sustainment, excessive age, natural disaster, fire, accident or other causes.
Sustainment	Maintenance and repair activities necessary to keep a typical inventory of facilities in “normal working condition”. Sustainment includes regularly scheduled maintenance as well as cyclical major repairs or replacement of components that occur periodically over the expected service life of the facilities.
Acronym	Title
AC	Alternating Current (electricity)
ANSI	American National Standards Institute
ASME	American Society of Mechanical Engineers
IMP	Integrated Maintenance Program
SRM	Sustainment, Restoration and Modernization
UFC	Unified Facilities Criteria

Attachment J-1602000-02
References and Technical Documents

Reference	Title
MO-209	Engineering Criteria
UFC 3-501-01	Electrical Engineering
UFC 3-550-01	Exterior Electrical Power Distribution
UFC 3-560-01	Electrical Safety. O&M
NAVFAC P-603	Utilities Infrastructure Management Manual
NAVFAC P-604	Electrical – Safe Acts for Employees (E-SAFE)
NETA MTS – 1997	Maintenance Testing Specifications for Electrical Power Distribution Equipment and Systems
NETA ATS – 1999	Maintenance Testing Specifications for Electrical Power Distribution Equipment and Systems
CEI 11-27	Requirement to work on LV (energized and not energized system), and HV system (no energized system)
CEI 78-17	Requirement for HV Maintenance to be comply at the CEI 0-16 (User connected at the distributor)
Non-Government Publications.	
American National Standards Institute, Inc. (ANSI)	

Attachment J-1602000-03 Electrical Power Transmission and Distribution System Description and Inventory

Reference Excel file “J-1602000-03 Electrical Power Transmission and Distribution Systems Description and Inventory”

Attachment J-1602000-04 Minimum Critical Spare Parts List
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Installation	Location	Equipment Type
Niscemi	Sub-Station	Sepam Relays S40
Niscemi	High Voltage System	24 kV Medium voltage Electric Cable
Niscemi	Sub-Station	20 kV HV SF6 Circuit Breaker
Critical spare parts are provided by the Contractor		

Estimated Minimum Stock Levels

Sepam Relays S40

Quantity: 2 units - for immediate replacement and backup for testing and maintenance.

24 kV Medium Voltage Electric Cable

Quantity: 50 meters minimum (or appropriate length based on installation scope) - sufficient length to replace or extend cables during maintenance or emergencies.

24 kV Medium Voltage Termination for Electric Cable

Quantity: 4 sets - to cover at least 2 transformers or feeders, including spares.

24 kV HV SF6 Circuit Breaker

Quantity: 2 units - operational and spare for maintenance or emergency replacement.

Thermostat and Temperature Control for Transformer Temperature

Quantity: 2 units - for immediate replacement and maintenance.

UPS for Substation Ancillary (substation-type with auto-recharge function)

Quantity: 2 units - primary unit and spare to ensure uninterrupted power supply during maintenance or failure.